UNIT-4

Cloud Fundamentals, Benefits of Cloud Computing, Type of Clouds, Cloud Computing Services, Cloud Computing Architecture, Virtualization and Cloud Computing, Grid Computing vs Cloud Computing, Security Concerns.

What is Cloud Computing?

Cloud Computing is defined as storing and accessing of data and computing services over the internet. It doesn't store any data on your personal computer. It is the on-demand availability of computer services like servers, data storage, networking, databases, etc. The main purpose of cloud computing is to give access to data centers to many users. Users can also access data from a remote server.

Benefits of Cloud Computing

Here, we will learn what are the benefits of Cloud Computing in your organization:

Cost Savings
Cloud computing removes the need for purchasing physical hardware and setting up infrastructure. The service provider manages maintenance, updates, and support, reducing operational expenses.

Strategic Edge
It gives businesses access to the latest software and technologies without heavy investment. This helps organizations innovate faster and stay ahead of competitors.

High Speed
Cloud services allow quick provisioning of computing resources within minutes. This reduces setup time and improves productivity.

Back-up and Restore Data
Cloud platforms provide automated backup solutions to protect data. Data recovery is faster and more efficient compared to traditional systems.

Automatic Software Integration
Applications in the cloud can be integrated easily and automatically. Users do not need complex manual configuration to connect services.

Reliability
Cloud providers ensure high availability and consistent performance. Regular updates and monitoring enhance system stability.

Mobility
Employees can access cloud services from any location using the internet. This supports remote work and flexible business operations.

Unlimited Storage Capacity
Cloud storage can be scaled up or down based on requirements. Businesses pay only for the storage space they actually use.

Collaboration
Multiple users can access and work on shared files simultaneously. Cloud platforms provide secure tools for real-time collaboration.

Quick Deployment
Cloud solutions can be implemented rapidly with minimal setup. The overall deployment time depends on the organization's needs and complexity.

Cloud Computing Architecture

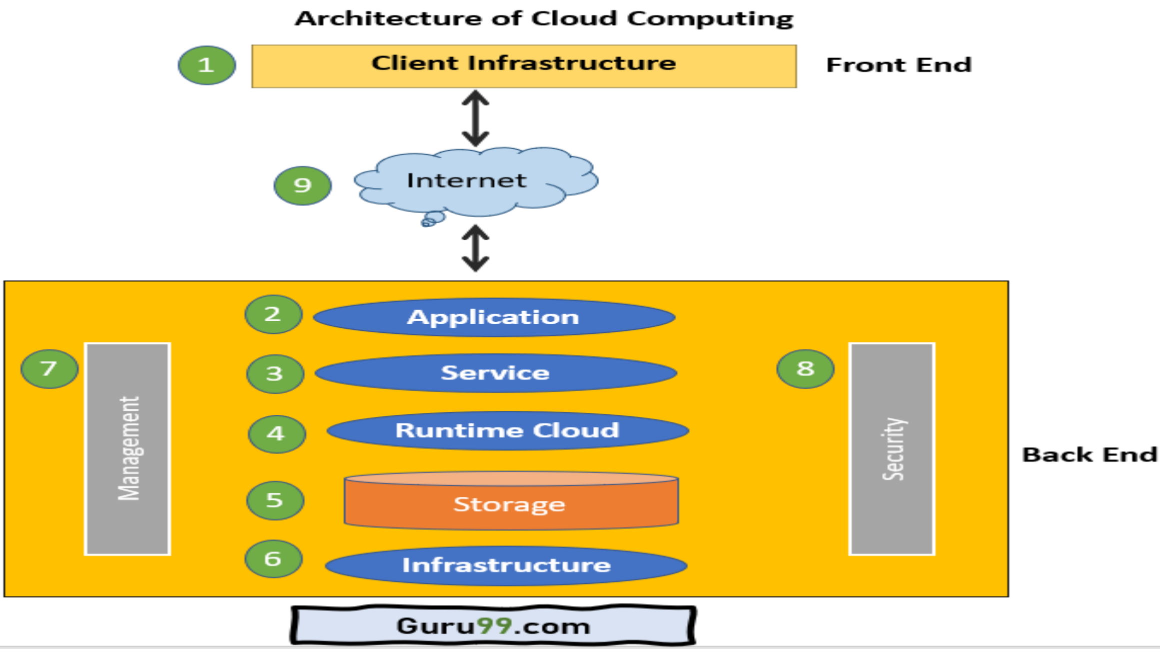
The Architecture of Cloud computing contains many different components. It includes Client infrastructure, applications, services, runtime clouds, storage spaces, management, and security. These are all the parts of a Cloud computing architecture.

Front End:

The client uses the front end, which contains a client-side interface and application. Both of these components are important to access the Cloud computing platform. The front end includes web servers (Chrome, Firefox, Opera, etc.), clients, and mobile devices.

Back End:

The backend part helps you manage all the resources needed to provide Cloud computing services. This Cloud architecture part includes a security mechanism, a large amount of data storage, servers, virtual machines, traffic control mechanisms, etc.



Important Components of Cloud Computing Architecture

Here are some important components of Cloud computing architecture:

1. Client Infrastructure

Client Infrastructure is a front-end component that provides a GUI. It helps users to interact with the Cloud.

2. Application

The application can be any software or platform which a client wants to access.

3. Service

The service component manages which type of service you can access according to the client's requirements.

Three Cloud computing services are:

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

4. Runtime Cloud

Runtime cloud offers the execution and runtime environment to the virtual machines.

5. Storage

Storage is another important Cloud computing architecture component. It provides a large amount of storage capacity in the Cloud to store and manage data.

6. Infrastructure

It offers services on the host level, network level, and application level. Cloud infrastructure includes hardware and software components like servers, storage, network devices, virtualization software, and various other storage resources that are needed to support the cloud computing model.

7. Management

This component manages components like application, service, runtime cloud, storage, infrastructure, and other security matters in the backend. It also establishes coordination between them.

8. Security

Security in the backend refers to implementing different security mechanisms for secure Cloud systems, resources, files, and infrastructure to the end-user.

9. Internet

Internet connection acts as the bridge or medium between frontend and backend. It allows you to establish the interaction and communication between the frontend and backend.

Benefits of Cloud Computing Architecture

Following are the cloud computing architecture benefits:

- Makes the overall Cloud computing system simpler.
- Helps to enhance your data processing.
- Provides high security.
- It has better disaster recovery.
- Offers good user accessibility.
- Significantly reduces IT operating costs.

Types of Cloud

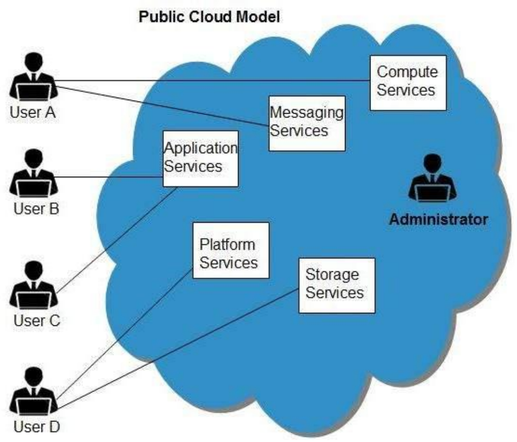
Let's take a closer look at the four main types of cloud computing, including

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud
4. Community Cloud.

Public Cloud
Public cloud is a model where a third-party provider owns and delivers computing resources over the internet, usually through a subscription model. It provides services like cloud applications, data storage, development environments, and virtual machines that allow users to create customizable systems and access them remotely.

It works on an on-demand basis, offering customized solutions while pooling resources to serve many users efficiently. This makes it suitable for businesses that need scalable and flexible infrastructure.

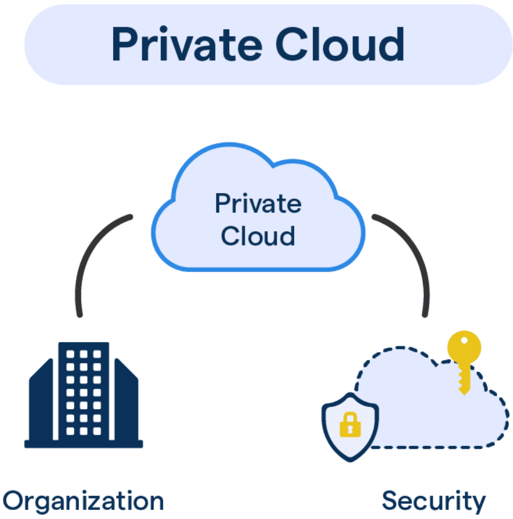
Pros	Cons
Easily scalable – resources can be increased or decreased as needed.	Limited control and visibility over backend infrastructure.
Cost-effective due to shared resources and flexible pricing.	Not suitable for highly resource-intensive applications requiring dedicated servers.



Private Cloud
Private cloud is a cloud computing model used exclusively by a single organization and not shared with others. It combines the control of on-premises infrastructure with cloud flexibility, allowing businesses to manage deployments, security, and updates internally or through a provider.

It is suitable for organizations with strict data privacy requirements and resource-intensive applications. It provides dedicated resources and greater control over the IT environment.

Pros	Cons
Full control over hardware, software, and security settings.	High initial cost for purchasing and setting up infrastructure.
No restrictions on applications and stronger data protection behind internal firewalls.	Complex setup and management due to multiple configuration options.

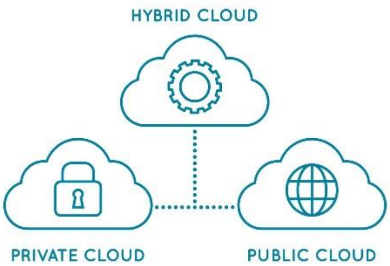


Hybrid Cloud

Hybrid cloud combines both public and private cloud infrastructures in a single environment. It allows organizations to keep sensitive data in a private cloud while using the public cloud for scalable workloads and SaaS applications.

Both environments can communicate and exchange data as required. It is widely used by businesses that need flexibility, cost control, and balanced security.

Pros	**Cons**
Flexible transition between public and private environments.	Can become complex and difficult to manage.
Scalable and cost-efficient while maintaining security requirements.	Private cloud portion can be expensive to maintain.

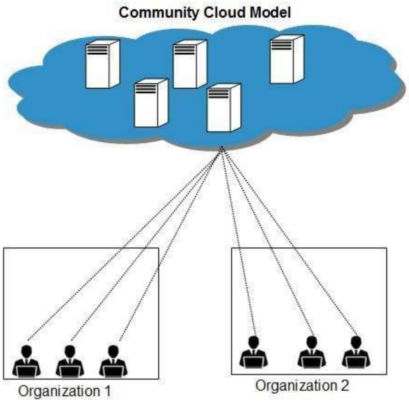


Community Cloud

Community cloud is a model designed for organizations with similar requirements, such as industry-specific security or compliance needs. It is similar to public cloud but customized to meet shared standards like data protection and regulatory policies.

It is commonly used in sectors such as finance, healthcare, insurance, and legal services. The provider configures infrastructure to match the community's specific operational and security needs.

Pros	**Cons**
Greater control and industry-specific customization.	Limited availability since it is a newer model.
Supports compliance needs and can integrate with hybrid setups.	More expensive than a traditional public cloud.



Parameters	**Public**	**Private**	**Community**	**Hybrid**
Setup and use	Easy	Need help from a professional IT team.	Require a professional IT team.	Require a professional IT team.
Scalability and Elasticity	Very High	Low	Moderate	High
Data Control	Little to none	Very High	Relatively High	High
Security and privacy	Very low	Very high	High	Very high

Parameters	Public	Private	Community	Hybrid
Reliability	Low	High	Higher	High
Demand for in-house software	No	Very high in-house software requirement	No	In-house software is not a must.

Grid vs cloud

What is Cloud Computing?

Cloud computing is a client–server architecture where data and applications are stored on remote servers and accessed through the internet. It provides resources like storage, software, and processing power as on-demand services.

Advantages (3)

1. Scalable resources based on demand.
2. Cost-effective with pay-as-you-use model.
3. Accessible from anywhere via internet.

Disadvantages (3)

1. Dependent on stable internet connectivity.
2. Security and privacy concerns.
3. Limited user control over infrastructure.

Grid computing

Grid computing is a distributed architecture where multiple computers work together to solve large computational tasks. Resources are shared collaboratively across different locations.

Advantages (3)

1. High resource utilization.
2. Supports parallel task processing.
3. Suitable for large scientific computations.

Disadvantages (3)

1. Complex to manage and maintain.
2. Security risks in distributed environment.
3. Limited flexibility compared to cloud.

Cloud Computing	Grid Computing
Client–server architecture.	Distributed architecture.
Centralized management.	Decentralized management.
Resources delivered as services.	Resources shared for computation.
Pay-as-you-use model.	Usually no direct usage cost.
Highly scalable.	Comparatively less scalable.
Accessed via web protocols.	Accessed via grid middleware.
Service-oriented (IaaS, PaaS, SaaS).	Application-oriented.
High accessibility.	Limited accessibility.
More flexible resource allocation.	Less flexible resource allocation.
Used for hosting apps and storage.	Used for complex scientific calculations.

Cloud Computing Services

The three main cloud service models are:

- **SaaS (Software as a Service)**
- **PaaS (Platform as a Service)**
- **IaaS (Infrastructure as a Service)**

Businesses use one or more of these based on their needs.

SaaS (Software as a Service)

SaaS is a software delivery model where applications are hosted by a provider and accessed over the internet. Instead of buying and installing software, users subscribe to it, usually monthly.

It works on any internet-enabled device and is used for tasks like accounting, sales, invoicing, and planning.

PaaS (Platform as a Service)

PaaS provides a cloud-based platform for developers to build, test, and deploy applications. It includes tools, storage, networking, and management services.

Developers focus on creating applications while the provider manages the infrastructure and updates.

IaaS (Infrastructure as a Service)

IaaS provides virtualized computing resources over the internet, such as virtual servers, storage, networks, and bandwidth. Resources are distributed across data centers for reliability.

It helps businesses reduce costs related to purchasing and maintaining physical IT infrastructure.

Virtualization and Cloud Computing

What is Virtualization?

Virtualization can be defined as a process that enables the creation of a virtual version of a desktop, operating system, network resources, or server. Virtualization plays a key and dominant role in cloud computing.

Virtualization as a Concept of Cloud Computing:

In cloud computing, Virtualization facilitates the creation of virtual machines and ensures the smooth functioning of multiple operating systems. It also helps create a virtual ecosystem for server operating systems and multiple storage devices, and it runs multiple operating systems.

Characteristics of Virtualization:

- **Distribution of resources:** Virtualization and Cloud Computing technology ensure end-users develop a unique computing environment. It is achieved through the creation of one host machine. Through this host machine, the end-user can restrict the number of active users.
- **Resource Isolation:** Virtualization provides isolated virtual machines. Each virtual machine can have many guest users.
- **Accessibility of server resources:** Virtualization delivers several unique features that ensure no need for physical servers.

Types of Virtualizations:

1)Application Virtualization-

This can be defined as the type of Virtualization that enables the end-user of an application to have remote access.

2) Network Virtualization-

This kind of virtualization can execute many virtual networks, and each has a separate control and data plan.

3) Desktop Virtualization

This can be defined as the type of Virtualization that enables the operating system of end-users to be remotely stored on a server or data center.

4) Storage Virtualization

This type of Virtualization provides virtual storage systems that facilitate storage management.

5) Server Virtualization-

This kind of Virtualization ensures masking of servers. The main or the intended server is divided into many virtual servers.

6) Data Virtualization-

This can be defined as the type of Virtualization wherein data are sourced and collected from several sources and managed from a single location.

Role of Virtualization in Cloud Computing:

Virtualization provides a virtual and isolated networking, storage, and memory area environment.

The Virtualization is achieved through a host machine and guest machine.

A host machine can be defined as the machine on which a virtual machine is developed, and the virtual machine so developed is termed as a guest machine.

Hardware virtualization plays a critical role by delivering infrastructure as a service solution most efficiently and effectively under a Cloud Computing process.

This type of Virtualization ensures portability.

Advantages of Virtualization:

- **It helps in cost reduction and boosting productivity towards the development process.**
- **It facilitates remote access to resources.**
- **It is highly flexible, and it allows the users to execute multiple desktops operating systems on one standard machine**
- **It removes the risks involved in terms of system failures**

Disadvantages of Virtualization:

- **The transition of the existing hardware setup to a virtualized setup requires an extensive time investment.**
- There is a lack of availability of skilled resources that helps in terms of transition of existing setup to virtual setup.
- **the implementation of Virtualization calls for high-cost implementations.**

Security concerns in cloud computing

1.Data Security & Privacy:

When you move to the cloud, all your company's sensitive information is handed over to a third-party provider — and hackers could potentially access this information.

2. DDoS Attacks & Performance Issues:

Since multiple businesses share the same server, a DDoS attack on one tenant can affect the performance of others on the same shared resource.

3.Downtime:

Cloud providers can face power loss, low internet connectivity, or service maintenance — all leading to downtime

4.Internet:

You can't access the cloud without an internet connection, and there is no other way to retrieve your data from the cloud.

5. Account Hijacking :

If login credentials of a user are stolen through phishing or other attacks, the hacker gets full access to that user's cloud account.

6. Insecure APIs:

Cloud services are accessed through APIs, and if these APIs are poorly designed or unprotected, it becomes easy for attackers to exploit and gain unauthorized access.